NATIONAL P. G. COLLEGE LUCKNOW

(An Autonomous College and CPE College of University of Lucknow)

DEPARTMENT OF COMPUTER SCIENCE

Syllabus for Four Year Undergraduate Programme

As per New Education Policy - 2020

Applicable from Session 2022-23 Onward

Programme Prerequisites:

- To study Computer Science, a student must have had the subject Mathematics learnt at 10+2 level with 50% minimum passing marks/grade (overall and in mathematics).
- Keen interest Computer Science & Application
- Skills and aptitude for scientific study and research
- Creativity and good comprehension while working on scientific procedures and research

Programme Introduction

Computer Science is the study of computers and technology. Computers have been shaping the future of mankind with the great surge in technologies like machine learning and IoT in the last decade. The curriculum of our subject aims to provide any pupil in the course to understand the architecture, theory, and math behind the technologies that drive our modern world forward.

B.Sc. in Computer Science facilitate the knowledge about the science behind computers and provide a platform to develop skills like programming, networking, and database administration. It also focuses on the ethics of developing and working with new technologies by providing strong arguments for green computing, security, and user privacy protection.

Programme outcomes (POs)	
PO 1	Gain a complete exposure to the theories and practices of Computer science.
PO 2	Get transformed into a skilled learner and active programmer, enabling the students to focus on their higher studies.
PO 3	Value computer professionals and programmers.
PO 4	Explore how the concepts and applications of Computer science lead to innovative thinking with a problem-solving attitude.
Programme specific outcomes (PSOs) Certificate	
PSO1	Develop and maintain problem-solving skills.
PSO2	Communicate Efficiently and with Confidence in English.
PSO3	Understand concepts of data organization.
PSO4	Solve trivial problems using programming languages.
Programme specific outcomes (PSOs) Diploma	
PSO1	Remember the aspects of behaviour people in an organization.
PSO2	Understand, create and maintain Relational Databases.
PSO3	Explore real-world problems, develop solutions using Computer.
PSO4	Familiarize with the importance of ethical hacking, its tool and ethical hacking process.
Programme specific outcomes (PSOs) Bachelor	
PSO1	Understand, analyse and develop computer programs in the areas related to algorithm, web design and networking for efficient design of computer-based system.
PSO2	To view the real-world problems from the spectacles of conceptual knowledge of Computer Science and to develop their solutions in a technical oriented way
PSO3	Apply standard software engineering practices and strategies in software project development using open-source programming environment to deliver a quality of product for business success.
PSO4	Work in the IT sector as system engineer, software tester, programmer, web developer

Semester-wise Titles of the Papers in B.Sc.

Year	Sem.	Course Code	Paper Title	Theory/Practical	Credits
Certificate in Science					
1	I	CSCC111T	Basics of Emerging Technologies	Theory	3
		CSCC112T	Operating System	Theory	3
	II	CSCC121T	Programming in C Language	Theory	3
		CSCC122P	Lab: Programming in C Language	Practical	3
Diploma in Science					
2	III	CSCC231T	Data Structures using C	Theory	3
		CSCC232P	Lab: Data Structures using C	Practical	3
	IV	CSCC241T	Digital Electronics	Theory	3
		CSCC242T	Data Communication and Computer Networks	Theory	3
Degree in Bachelor of Science					
3	V	CSCC351T	Database Management Systems	Theory	4
		CSCC352T	Software Engineering	Theory	4
		CSCC353P	Lab: Database Management Systems	Practical	4
	VI	CSCC361T	Problem solving using Python	Theory	4
		CSCC362T	E Commerce	Theory	4
		CSCC363P	Lab: Problem solving using Python	Practical	4
Bachelor (Research) in Science					
4	VII	CSCC471T	Research Methodology	Theory	4
		CSCC472T	Cyber Security	Theory	4
		CSCC473T	Artificial Intelligence	Theory	4
		CSCC474T	Web Designing	Theory	4
		CSCC475P	Lab: Web Designing	Practical	4
		CSCC476I	Internship	Internship	4
	VIII	CSCC481T	Data Analytics	Theory	4
		CSCC482T	Software Testing	Theory	4
		CSCC483T	Web Development	Theory	4
		CSCC484P	Lab: Web Development	Practical	4
		CSCC485R	Research Project	Project	12

Semester wise Syllabus of Computer Science in B.Sc.

Programme/Class: **Certificate**	Year: **First**	Semester: **First**
Subject: **Computer Science**		
Course Code: **CSCC111T**	Course Title: **: Basics of Emerging Technologies**	

Course outcomes:

- To understand Functional Components of Computers.
- Effectively explains the basic concepts of Hardware: Input and out- put devices.
- To understand Software and its types
- Able to understand Operating system, Functions of Operating System.
- Able to get overview of Data Communication and Internet

Credits: 3	**Core Compulsory**
Max. Marks: **30+70**	
Total No. of Lectures-Tutorials-Practical (in hours per week): **3-0-0**	

Unit	Topic	No. of Lectures
I	Introduction, Characteristics of Computers, Block diagram of computer, Types of computers and features: Mini Computers, Micro Computers, Mainframe Computers, Super Computer. Types of Programming Languages: Machine Languages, Assembly Languages, High Level Languages. Translators:- Assembler ,Compiler, Interpreter and Linkers, Operating system concepts, Types of OS, Functions of OS	10
II	I/O Devices:- Keyboard, Mouse, Scanner, Light pen, Trackball, Joystick, Bar NUMBER reader, OCR, OMR, MICR, Digitizer; Monitor, Printer, Plotter etc., Memory concepts, Types of Memory (Primary and Secondary):-RAM,ROM and its types, Secondary Storage Devices (Magnetic tape, Magnetic Disk(FD, HD),Optical Disk(CD,DVD), Pen drive), Data Organization: Drives, Files, Directories	12
III	Basic elements of a communication system, Data transmission modes, Data Transmission speed, Data transmission media, Digital and Analog Transmission, Network topologies, Network Types (LAN, WAN and MAN), Client and Servers, Intranet, Extranet. **Internet:** Introduction to Internet, Terminologies related to Internet: Protocol, Domain name, IP address, URL, World Wide Web, Connecting to the Internet	11
IV	Introduction to Big Data: Types of digital data, Big Data architecture and characteristics, Big Data technology components, Big Data importance and applications, Introduction to Block Chain: Structure, Operational aspects of Bitcoin Block, Compare different types of Block chains, Protocols, Payment mode code execution.	12

Suggested Readings:

- P. K. Sinha & Priti Sinha , "Computer Fundamentals", BPB Publications, 2007.
- Dr. Anita Goel, Computer Fundamentals, Pearson Education, 2010.
- Fundamental of Computers – By V. Rajaraman B.P.B. Publications

Suggested online links:

- https://onlinecourses.swayam2.ac.in/cec19_cs06/preview
- https://www.youtube.com/watch?v=XZb1uzSGOmU
- https://www.pearsoned.co.in/prc/book/anita-goel-computer-fundamentals-1e1/9788131733097

Programme/Class: **Certificate**	Year: **First**	Semester: **First**
Subject: **Computer Science**		
Course Code: **CSCC112T**	Course Title: **Operating System**	

Course outcomes: After the completion of the course the students will be able:

- Understand role, responsibilities, features, and design of operating system.
- Analyse memory management schemes and process scheduling algorithms.
- Apply process synchronization techniques to formulate solution for critical section problems.
- Illustrate concept of disk scheduling.
- Evaluate process deadlock handling techniques.

Credits: **3**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 3-0-0

Unit	Topic	No. of Lectures
I	**Introduction** Operating system and functions: Process Management, Device Management, Memory Management, Information Management, Classification of Operating systems: Batch, Interactive, Multiprogramming, Time sharing, Real Time System, Multiprocessor Systems, Parallel System, Distributed System. Multiuser Systems, Multithreaded Systems and real time System Operating System Structure, System Components, Operating System Services.	9
II	**Process Management** Process Management: Processes: Definition, Process Relationship, Process states, Process State transitions, Process Control Block, Context switching – Threads, Interprocess Communication, Critical Section. Process Scheduling: Definition, Scheduling objectives, Types of Schedulers, Long, Medium, Short Scheduling. Scheduling criteria: CPU utilization, Throughput, Turnaround Time, Waiting Time, Response Time (Definition only). Pre-emptive and Non Pre-emptive Scheduling. Scheduling Algorithms: First-Come-First-Serve, Shortest Job First, Shortest Remaining Time Next, Round Robin, Priority Scheduling, DeadLock: Resource allocation graph, Introduction: Characterization, Deadlock conditions, Dead lock prevention, Avoidance and Detection. Safe state condition.	13
III	**Memory Management** Memory allocation, Relocation, Protection, Sharing, Logical and Physical address space, Swapping, Contiguous allocation space, Fragmentation: Internal and External. Paging, Segmentation, Virtual Memory Management: Demand paging, Thrashing. Page Replacement Algorithms,: FIFO, LRU, OPR	11
IV	**Information Management and Disk Scheduling** Disk storage and disk **:** File concept, , and File sharing, File system implementation issues, File system protection and security. scheduling, File System: File concepts, File organization and access mechanism, File directories, Directory Structure File-System Implementation: File system structures, Directory Implementation, Allocation methods. Disk	12

	Management: Disk Structure, Free space Management. Disk Scheduling Algorithm: FCFS, SSTF, SCAN, LOOK, C-SCAN, C-LOOK	

Suggested Readings:

- Andrew S. Tanenbaum and Herbert Bos,"Modern Operating Systems," Fourth Edition, Pearson, 2014.
- Abraham Silberschatz, Greg Gagne, and Peter B. Galvin, "Operating System Concepts," Tenth Edition, Wiley, 2018.
- William Stallings, "Operating Systems: Internals and Design Principles," Seventh Edition, Prentice Hall, 2011.
- Dhanjay Dhamdhere, "Operating Systems," First Edition, McGraw-Hill, 2008
- Milan Milankovic "Operating systems, Concepts and Design" McGraw Hill

Suggested online links:

- https://nptel.ac.in/courses/106/105/106105214/

Programme/Class: **Certificate**	Year: **First**	Semester: **Second**
Subject: **Computer Science**		
Course Code: **CSCC121T**	Course Title: **Programming in C Language**	

Course outcomes: On completion of the course, the student will be able to:

- Use the fundamentals of C programming in trivial problem solving.
- Enhance skill on problem solving by constructing algorithms.
- Identify solution to a problem and apply control structures and user defined functions for solving the problem.
- Apply skill of identifying appropriate programming constructs for problem solving.

Credits: **3**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 2-1-0

Unit	Topic	No. of Lectures
I	Programming Techniques and Concepts, Overview of C, History, Structure of a C program, Compiling a C program and executing a C program , C character set, Keywords in C, Operators: Arithmetic, unary, logical, bitwise, assignment and conditional operators and Hierarchy of operators, Basic data types, Qualifiers used with basic data types, Data Types Declarations, Constants and Variables, Expression and statement and symbolic constants, Basic I/O function, Preprocessor command: #include, define.	10
II	Control statements: if statement, if else statement, Construction of loops and implementation, control statement: for statement, while loop, do while statements, break statements, continue statements, Nested loops, switch statement, goto statement, ternary operators. Library Functions.	10
III	Array: Advantages of arrays, one dimensional and two dimensional arrays, types of arrays, array declaration, array initialization, accessing data from array, array inside the memory. Character arrays, Array overflow, Null terminated Strings as Array of Characters, String Variables, Reading & writing strings, string handling functions. Pointer: Declaration, uses of pointers, array of pointers, Passing pointer to a Function, Operations on Pointers. Array of pointers, Pointer and Strings, Storage classes: automatic, external and static variables, Functions: functions prototypes, function call, call by value, call by reference, Recursion, Advantages of functions, declaring a function, calling a function, variables, passing arguments to a function, nested functions, passing array to functions, recursion in functions, Call by value and Call by reference.	15
IV	Structure variables, Initialization, Structure Assignment, Nested Structures, Structures and Functions, Structures and Arrays. Union: Definition and declaration; Differentiate between Union and structure. Introduction to File Handling: File structure, File handling function, File types, Streams, Text, Binary, File system basics, Unformatted file processing: The file pointer, Opening a file, reading from file, writing to file, Closing a file.	10

Suggested Readings:

- Brian W. Kernighan and Dennis Ritchie, “The C Programming Language” Pearson Publication
- Let us C-Yashwant Kanetkar
- K.R.Venugopal, S.R.Prasad, “Mastering C” McGraw-Hill Education India

Suggested online links:

- https://nptel.ac.in/noc/courses/noc22/SEM1/noc22-cs40/
- http://www.learn-c.org/
- http://crasseux.com/books/ctutorial/

Programme/Class: **Certificate**	Year: **First**	Semester: **Second**
Subject: **Computer Science**		
Course Code: **CSCC122P**	Course Title: **Lab: Programming in C Language**	

Course outcomes: On completion of the course, the student will be able to:

- Use the fundamentals of C programming in trivial problem solving.
- Enhance skill on problem solving by constructing algorithms.
- Identify solution to a problem and apply control structures and user defined functions for solving the problem.
- Apply skill of identifying appropriate programming constructs for problem solving.

Credits: **3**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 0-0-3

Topic

PART A:

1.Write a C Program to read radius of a circle and to find area and circumference
2. Write a C Program to read three numbers and find the biggest of three
3. Write a C Program to demonstrate library functions in math.h
4. Write a C Program to check for prime
5. Write a C Program to generate n primes
6. Write a C Program to read a number, find the sum of the digits, reverse the number and check it for palindrome
7. Write a C Program to read numbers from keyboard continuously till the user presses 999 and to find the sum of only positive numbers
8. Write a C Program to read percentage of marks and to display appropriate message (Demonstration of else-if ladder)
9. Write a C Program to find the roots of quadratic equation (demonstration of switch-case statement)
10. Write a C program to read marks scored by n students and find the average of marks (Demonstration of single dimensional array
11. Write a C Program to remove Duplicate Element in a single dimensional Array
12. Program to perform addition and subtraction of Matrices

PART B:

1. Write a C Program to find the length of a string without using built in function
2. Write a C Program to demonstrate string functions.
3. Write a C Program to demonstrate pointers in C
4. Write a C Program to check a number for prime by defining isprime( )function
5. Write a C Program to read, display and to find the trace of a square matrix
6. Write a C Program to read, display and add two m x n matrices using functions
7. Write a C Program to read, display and multiply two m x n matrices using functions
8. Write a C Program to read a string and to find the number of alphabets, digits, vowels, consonants, spaces and special characters.
9. Write a C Program to Reverse a String using Pointer
10. Write a C Program to Swap Two Numbers using Pointers
11. Write a C Program to demonstrate student structure to read & display records of n students.
12. Write a C Program to demonstrate the difference between structure & union.

Programme/Class: **Diploma**	Year: **Second**	Semester: **Third**
Subject: **Computer Science**		
Course Code: **CSCC231T**	Course Title: **Data Structures using C**	

Course Outcomes:

- Understand data structure concepts, characteristics, describe data organization schemes such as arrays, stacks and their applications.
- To learn about queue data structure, types of queue, operations on queue.
- Familiarize with concepts of linked lists and operations on them.
- Learn about non-linear data-structure tree, its terminologies, tree traversal and other operations.
- Learn about binary search tree, B-tree working and its application.
- Learn and understand various searching and sorting techniques.

Credits: **3**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 2-1-0

Unit	Topic	No. of Lectures
I	Data Structure, definition, and application, Arrays: Representation of single and multidimensional arrays; sparse arrays - lower and upper Triangular matrices and Tri-diagonal matrices, Recursion, Dynamic memory allocation, malloc () function, Size of () operator, Function free (), Function realloc(). Defining Stack and Queue, Stack Operations and Implementation, Array Implementation, Pointer Implementation, Stack Applications, Convert Number Bases by Using Stacks, Infix to Postfix Conversion, Evaluation of postfix expression. Introduction and primitive operation on queues, D-queues Queue Application, Circular, Double Ended and Priority Queues.	10
II	Lists, Basic Terminology, Static Implementation of Lists, Pointer Implementation of Lists, Insertion in a list, Deletion from a list, traversal, Searching, Arrays using Linked List, Doubly Linked Lists, Circular, Doubly, Circular Doubly Linked List	12
III	Defining Graph, Basic Terminology, Graph Traversal: Depth First Search (DFS), Breadth First Search (BFS), Shortest Path Problem,Trees: Introduction and terminology; Traversal of binary trees; Algorithms for tree operations such as traversal, insertion, deletion; Binary Trees, Inorder, Postorder and Preorder Traversals, Minimal Spanning Tree, Prims and Kruskals Algorithm, Binary Search Trees, Operations on a BST, Insertion, Deletion , Search for a key in BST, B-Trees: Introduction: Threaded Binary tree, AVL Trees	12
IV	Searching and Sorting techniques, Sequential Search, Binary Search, Selection Sort, Insertion Sort, Bubble Sort, Quick Sort, 2-way Merge Sort, Heap Sort, Bucket or Radix sort. Hashing.	11

Suggested Readings:

- Data Structures- Seymour Lipschutz

- Data Structures using C and C++- Tanenbaum

Suggested online links:

- https://nptel.ac.in/courses/106/102/106102064/
- https://nptel.ac.in/courses/106/106/106106127/

Programme/Class: **Diploma**	Year: **Second**	Semester: **Third**
Subject: **Computer Science**		
Course Code: **CSCC232P**	Course Title: **Lab: Data Structures using C**	

Course Outcomes:

- Understand data structure concepts, characteristics, describe data organization schemes such as arrays, stacks and their applications.
- To learn about queue data structure, types of queue, operations on queue.
- Familiarize with concepts of linked lists and operations on them.
- Learn about non-linear data-structure tree, its terminologies, tree traversal and other operations.
- Learn about binary search tree, B-tree working and its application.
- Learn and understand various searching and sorting techniques.

Credits: **3**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 0-0-3

Practical List of Data Structures with C:

- Write a program that uses functions to perform the following:
- Create a singly linked list of integers.
- Delete a given integer from the above linked list.
- Display the contents of the above list after deletion.
- Write a program that uses functions to perform the following:
- Create a doubly linked list of integers.
- Delete a given integer from the above doubly linked list.
- Display the contents of the above list after deletion.
- Write a program that uses stack operations to convert a given infix expression into its postfix Equivalent, implement the stack using an array.
- Write program to implement a double ended queue using
- i) array and ii) doubly linked list respectively.
- Write a program that uses functions to perform the following:
- Create a binary search tree of characters.
- Traverse the above Binary search tree recursively in Postorder.
- Write a program that uses functions to perform the following:
- Create a binary search tree of integers.
- Traverse the above Binary search tree non recursively in inorder.
- Write program for implementing the following sorting methods to arrange a list of integers in ascending order:
- Insertion sort
- Merge sort
- Write program for implementing the following sorting methods to arrange a list of integers in ascending order:
- Quick sort
- Selection sort
- Write program to implement Insertion Sort (The program should report the number of comparisons)
- Write program implement Merge Sort(The program should report the number of comparisons)
- Write program implement Heap Sort (The program should report the number of comparisons)
- Write program implement Randomized Quick sort (The program should report the number of comparisons)
- Write program for creation and traversal of Binary Search Tree.

Programme/Class: **Diploma**	Year: **Second**	Semester: **Fourth**
Subject: **Computer Science**		
Course Code: **CSCC241T**	Course Title: **Digital Electronics**	

Course Outcomes:

- Understand Digital Computer and Digital Systems.
- Understand the logic and applications of Boolean algebra and logic gates.
- Understand the concept of Combinational circuits, Sequential circuits and memory

Credits: **3**	**Core Compulsory**
Max. Marks: **30+70**	
Total No. of Lectures-Tutorials-Practical (in hours per week): 3-0-0	

Unit	Topic	No. of Lectures
I	Binary System : Digital Computers and digital systems , Number system, Binary Arithmetic , Signed binary numbers, complements : r's complements(2's complement, 10's complements), (r-1)'s complements (1's complement, ,9's complement) , Subtraction using 1's complement and 2's complement method , Binary NUMBERs ,Logic gates: Inverter, AND, OR, NOR, NAND, XOR, XNOR, De-Morgan's Theorems, Boolean algebra, Canonical and standard forms , Karnaugh Map , Introduction to IC Digital logic families(RTL, DTL, TTL, ECL, MOS and CMOS) and characteristics (Fan-In ,Fan-out, Power Dissipation , Propagation delay ,Noise margin) of IC Digital logic families (TTL , ECL , CMOS)	10
II	Combinational and sequential Circuits : Half adder , Full adder , Half Subtractor , full Subtractor , Serial and parallel adder, NUMBER conversion circuits , Parity generator and checker, Comparators , Encoder , Decoder , Multiplexer , De-multiplexer	12
III	Flip flops: RS-Flip flop, D-Flip flop, T-flip flop, JK-Flip flop , JK Master slave flip flop, edge triggered and pulse triggered flip flops, Registers: Simple Register , Shift registers ,Types of Shift Register, Construction and timing diagram of 4-bit Shift Register	11
IV	**Counter** - Binary counter, characteristic of counter-(Synchronous/Asynchronous, Modulus/length of counter, UP/DOWN counter, Speed of Asynchronous counter) Construction of full length and Non full length, UP/DOWN Synchronous counter using 2, 3 and 4 FF, Ripple counter. Design of different MOD counter, Construction of Asynchronous counter, Ring Counter, Johnson Counter, Random Access Memory, Read Only Memory, Types of Read Only Memory, Multivibrators-Astable, Monostable, Bistable, Schmitt Trigger, Timer	12

Suggested Readings:

- "Modern Digital Electronics" - R.P. Jain
- Digital logic and Computer design- M. Morris Mano

Suggested online links:

- https://nptel.ac.in/courses/108/105/108105132/

Programme/Class: **Diploma**	Year: **Second**	Semester: **Fourth**
Subject: **Computer Science**		
Course Code: **CSCC242T**	Course Title: **Data Communication and Computer Network**	

Course outcomes:

- After the completion of the course the students will be able:
- To develop understanding of computer networks and communication basics.
- To understand design issues and services at different layers of reference models.
- To learn various error detection/correction techniques, routing protocols, congestion control algorithms, and connection establishment/release.
- To describe and analyse related technical, administrative, and social aspects of networking.

Credits: **3**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 3-0-0

Unit	Topic	No. of Lectures
I	Computer networks, Networks Hardware-Local Area network, Metropolitan Area network, Wide Area network, Wireless network, Internetworks, Networks Software: Protocol Hierarchies, Design and Issues for layers, Interfaces and Services, Connection oriented and Connection less Services	10
II	Reference Models, and OSI Reference Model, TCP/IP Protocol Suit The Physical Layer: Maximum Data Rate of a channel, Transmission Media: Magnetic Media, Twisted Pair, Baseband and Broadband Coaxial cable, Fibre Optical Wireless Transmission, structure of telephone system, Switching, Multiplexing, ISDN	12
III	Data link layer, Error control, Flow control, Sliding Window Protocol, Channel Allocation Problem, Multiple Access Protocol: ALOHA, CSMA protocol, Collision Free protocol, Polling, Network layer: Routing Algorithm, Congestion Control Algorithm, IP protocol, IP Addresses.	12
IV	Transport Layer: Addressing, Establishing and releasing a connection, TCP service Model, TCP protocol, the Application Layer: Network Security, Domain Name System	11

Suggested Readings:

- B. A. Forouzan: Data Communications and Networking, Fourth edition, THM ,2007
- S. Tanenbaum: Computer Networks, Fourth edition, PHI, 2002
- James F. Kurose, Keith W. Ross, “Computer Networking”, Pearson Education.
- Michael A. Gallo, William M. Hancock, “Computer Communications and Networking Technologies”, CENGAGE Learning.

Suggested online links:

- https://nptel.ac.in/noc/courses/noc22/SEM1/noc22-cs19/

Programme/Class: **Degree**	Year**: Third**	Semester: **Fifth**
Subject**: Computer Science**		
Course Code: **CSCC351T**	Course Title: **Database Management System**	

Course outcomes:

- After the completion of the course the students will be able to:
- Understands the basic concepts of data base management systems.
- Design E-R diagrams for real world applications.
- Formulate relational algebraic expressions using relational data models and languages.
- Apply normalization transaction properties and concurrency control to design database.
- Analyze the security algorithms for database protection.

Credits: **4**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 3-1-0

Unit	Topic	No. of Lectures
I	Data, Information and knowledge, introducing database and different kinds of database users, concept or a database, interacting with a database, architecture of a database, using relational databases, basics of relational databases, using relational databases, identifiers for relations, characteristic of database, database system concepts and data independence, content of data dictionary, data administration function.	15
II	Traditional data model – ANSI/SPRC, 3-level architecture, over view, of three traditional models – hierarchical, network and relational models, comparison of these models, ER model, File organization technique – random, file organization technique, multi key file organization technique, Indexing, concurrency control, database security, database recovery.	15
III	Data Normalization Introduction to SQL: Characteristics and advantages, SQL Data Types and Literals, DDL, DML, SQL Operators, Tables: Creating, Modifying, Deleting, Views: Creating, Dropping, Updating using Views SQL DML Queries: SELECT Query and clauses, Set Operations, Predicates and Joins, Aggregate Functions, Nested Queries.	15
IV	Specifying constraints and indexes in SQL, data manipulation, multiple table operations. Database Modification using SQL Insert, Update and Delete, T-SQL: Cursor, concept of Stored Procedures, Functions and Triggers.	15

Suggested Readings:

- Henry F. Korth and Abraham Silberschatz, "Database System Concepts," Second Edition,McGraw Hill, 1991.
- AtulKahate, "Introduction to Database Management Systems," Pearson India, 2004.
- Raghu Ramakrishnan and Johannes Gehrike, "Database Management Systems," ThirdMcGraw Hill, Edition, 2003.
- R. Elmasri, S.B. Navathe Database Systems Models, Languages, Design and application Programming, 6 Edition, Pearson Education,2013.
- A. Silberschatz, H.F. Korth, S. Sudarshan, Database System Concepts 6th Edition,McGraw Hill, 2010.
- C.J Date " An Introduction to Database Systems", Addison Wesley

Suggested online links:

- https://nptel.ac.in/noc/courses/noc22/SEM1/noc22-cs57/

Programme/Class: **Degree**	Year: **Third**	Semester: **Fifth**
Subject: **Computer Science**		
Course Code: **CSCC352T**	Course Title: **Software Engineering**	

Course Outcomes:

- After the completion of the course the students will be able to:
- Explain various software characteristics and analyze different software Development Models. Understands the basic concepts of data base management systems.
- Demonstrate the contents of a SRS and apply basic software quality assurance practices to ensure that design, development meet or exceed applicable standards.
- Formulate relational algebraic expressions using relational data models and languages.
- Compare and contrast various methods for software design.
- Formulate testing strategy for software systems, employ techniques such as unit testing, Test driven development and functional testing.

Credits: **4**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 4-0-0

Unit	Topic	No. of Lectures
I	**Software Engineering Fundamentals: Definition** of software product and process, Software Crisis, Software development paradigms, Software lifecycle models: Waterfall Model, Prototyping Model, Iterative Enhancement Model, Evolutionary Development Model and Spiral Model.	15
II	**Software Requirement Analysis & Specification: System** specification, Software requirements specification (SRS) standards, Formal specification methods, Specification tools, Requirements validation and management. **Software Process:** Software Process and Models, Tools and Techniques of Process Modelling, Product and Process. Important qualities of software product and process: correctness, reliability, robustness, user friendliness, verifiability, maintainability, reusability, portability, data abstraction, modularity, Principles of software engineering.	15
III	**Software Design:** Software architecture, Modular design - cohesion and coupling, Process-oriented design, Data-oriented design, User-interface design, Real-time software design.**CASE Tools:** Relevance of CASE tools, High-end and low-end CASE tools, Automated support for data dictionaries, DFD, ER diagrams.	15
IV	**Coding and Testing:** Choice of Programming languages, Coding standards, Introduction to Testing Process, Functional & Structural Testing, Testing Integration & System Testing, Testing tools.	15

Suggested Readings:

- RS Pressman, Software Engineering: A Practitioners Approach, McGraw Hill.
- Pankaj Jalote, Software Engineering, Wiley
- Rajib Mall, Fundamentals of Software Engineering, PHI Publication.
- KK Aggarwal and Yogesh Singh, Software Engineering, New Age International Publishers.
- Ghezzi, M. Jarayeri, D. Manodrioli, Fundamentals of Software Engineering, PHI Publication.
- Ian Sommerville, Software Engineering, Addison Wesley.
- Kassem Saleh, "Software Engineering", Cengage Learning.
- P fleeger, Software Engineering, Macmillan Publication

Suggested online links:

- https://nptel.ac.in/courses/106/105/106105182/

Programme/Class: **Degree**	Year: **Third**	Semester: **Fifth**
Subject: **Computer Science**		
Course Code: **CSCC353P**	Course Title: **Lab: Database Management Systems**	

Course outcomes: Ability to:

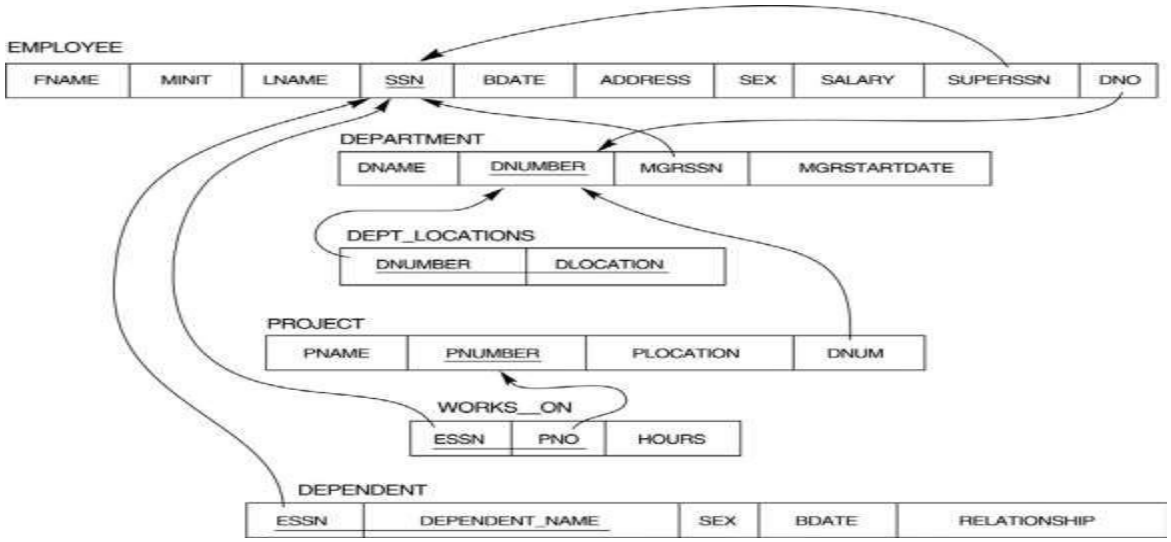
- Understand, analyse and apply common SQL statements including DDL, DML and DCL statements to perform different operations.
- Design and implement a database schema for a given problem.

Credits**: 4**	**Core Compulsory**
Max. Marks: **30+70**	
Total No. of Lectures-Tutorials-Practical (in hours per week): 0-0-4	

List of Experiments

- Creation of databases and execution of SQL queries.
- Creation of Tables using MySQL: Data types, Creating Tables (along with Primary and Foreign keys), Altering Tables and Dropping Tables.
- Practicing DML commands- Insert, Select, Update, Delete.
- Practicing Queries using ANY, ALL, IN, EXISTS, NOT, EXISTS, UNION, INTERSECT, and CONSTRAINTS, etc.
- Practice Queries using COUNT, SUM, AVG, MAX, MIN, GROUP BY, HAVING, VIEWS Creation and Dropping.
- Use of COMMIT, ROLLBACK and SAVEPOINT.
- Practicing on Triggers - creation of trigger, Insertion using trigger, Deletion using trigger, Updating using trigger.
- To remove the redundancies and anomalies in the above relational tables, Normalize up to Third Normal Form.

Relational Database Schema - COMPANY

1. 

2. Questions to be performed on above schema
3. Create tables with relevant foreign key constraints
4. Populate the tables with data
5. Perform the following queries on the database :
    a. Display all the details of all employees working in the company.
    b. Display ssn, lname, fname, address of employees who work in department no 7. Retrieve the birthdate and address of the employee whose name is 'Franklin T. Wong'
    c. Retrieve the name and salary of every employee
    d. Retrieve all distinct salary values
    e. Retrieve all employee names whose address is in ‘Bellaire’

f. Retrieve all employees who were born during the 1950s
g. Retrieve all employees in department 5 whose salary is between 50,000 and 60,000(inclusive)
h. Retrieve the names of all employees who do not have supervisors
i. Retrieve SSN and department name for all employees
j. Retrieve the name and address of all employees who work for the 'Research' department
k. For every project located in 'Stafford', list the project number, the controlling department number, and the department manager's last name, address, and birthdate.
l. For each employee, retrieve the employee's name, and the name of his or her immediate supervisor.
m. Retrieve all combinations of Employee Name and Department Name
n. Make a list of all project numbers for projects that involve an employee whose last name is 'Narayan' either as a worker or as a manager of the department that controls the project.
o. Increase the salary of all employees working on the 'ProductX' project by 15%. Retrieve employee name and increased salary of these employees.
p. Retrieve a list of employees and the project name each works in, ordered by the employee's department, and within each department ordered alphabetically by employee first name.
q. Select the names of employees whose salary does not match with salary of any employee in department 10.
r. Retrieve the name of each employee who has a dependent with the same first name and same sex as the employee.
s. Retrieve the employee numbers of all employees who work on project located in Bellaire, Houston, or Stafford.
t. Find the sum of the salaries of all employees, the maximum salary, the minimum salary, and the average salary. Display with proper headings.
u. Find the sum of the salaries and number of employees of all employees of the 'Marketing' department, as well as the maximum salary, the minimum salary, and the average salary in this department.
v. Select the names of employees whose salary is greater than the average salary of all employees in department 10.
w. For each department, retrieve the department number, the number of employees in the department, and their average salary.
x. For each project, retrieve the project number, the project name, and the number of employees who work on that project.
y. Change the location and controlling department number for all projects having more than 5 employees to 'Bellaire' and 6 respectively.
z. For each department having more than 10 employees, retrieve the department no, no of employees drawing more than 40,000 as salary.
aa. Insert a record in Project table which violates referential integrity constraint with respect to Department number. Now remove the violation by making necessary insertion in the Department table.
bb. Delete all dependents of employee whose ssn is '123456789'.
cc. Delete an employee from Employee table with ssn = '12345'(make sure that this employee has some dependents, is working on some project, is a manager of some department and is supervising some employees). Check and display the cascading effect on Dependent and Works on table. In Department table MGRSSN should be set to default value and in Employee table SUPERSSN should be set to NULL
dd. Perform a query using alter command to drop/add field and a constraint in Employee table.

Suggested Readings:

- Paul DuBois, "MySQL Cookbook: Solutions for Database Developers and Administrators," Third Edition, O'Reilly Media, 2014.
- Frank M. Kromann, "Beginning PHP and MySQL: From Novice to Professional," Fifth Edition, Apress, 2018.
- Joel Murach and Ray Harris, "Murach's PHP and MySQL," First Edition, Mike Murach & Associates, 2010.
- Luke Welling, Laura Thomson, "PHP and MySQL Web Development," Fourth Edition, Addison-Wesley, 2008.

Programme/Class: **Degree**	Year: **Third**	Semester: **Sixth**
Subject: **Computer Science**		
Course Code: **CSCC361T**	Course Title: **Problem solving using Python**	

Course outcomes: After the completion of the course the students will be able to:

- Understand hardware components of computer system such as memory system organization, input/output devices, aware of software components of computer system, and windows operating system concepts.
- Develops basic understanding of computers, the concept of algorithm and algorithmic thinking.
- Develops the ability to analyze a problem, develop an algorithm to solve it.
- Develops the use of the Python programming language to implement various algorithms, and develops the basic concepts and terminology of programming in general.

Credits: **4**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 3-1-0

Unit	Topic	No. of Lectures
I	**Introduction to Python:** Features, Installation, setting up path and Working with Python Understanding Python Variables Python Basic Operators Understanding python blocks, Python Data Types: Declaring and using Numeric data types: int, float, complex Using string data type and string operations Defining list and list slicing Use of Tuple data type. Python Program Flow Control Conditional blocks using if, else and nested if-else Simple for loops in python for loop using ranges, string, list and dictionaries Use of while loops in python, Nested loop, Loop manipulation using pass, continue, break and else.	15
II	**String Manipulation:** Accessing Strings, Basic Operations, String slices. Lists: Introduction, accessing list, Operations, working with lists. Function and methods, Tuples: Accessing Tuples, Operations, working, Function and methods. Dictionaries: Accessing values in Dictionaries, working with dictionaries, properties, functions.	15
III	**Functions:** Defining a function, calling a function, Types of functions, function arguments, Anonymous functions, Global and local variables. Modules: Importing modules, Math module, Random module, Packages, Composition.	15
IV	**Input Output:** Printing on screen, reading data from keyboard, opening and closing files, reading and writing files, functions. Exception Handling. OOPs concept: Class and object, Attributes, Inheritance, Overloading, overriding, Data hiding, Regular Expressions.	15

Suggested Readings:

- T. Budd, Exploring Python, TMH, 1st Ed, 2011
- Allen Downey, Jeffrey Elkner, Chris Meyers , How to think like a computer scientist : learning with Python , Freely available online.2012
- Rober Sedgewick, K Wayne -Introduction to Programming in Python: An interdisciplinary Approach" Pearson India

Suggested online links:

- http://docs.python.org/3/tutorial/index.html
- http://interactivepython.org/courselib/static/pythonds
- http://www.ibiblio.org/g2swap/byteofpython/read/

Programme/Class: **Degree**	Year: **Third**	Semester: **Sixth**
Subject: **Computer Science**		
Course Code: **CSCC362T**	Course Title: **E-Commerce**	

Course outcomes:
After the completion of the course the students will be able to:
Understands the basic concepts of E-Commerce.
Understand about firewall and EDI.
Understand about Electronic Payment System.
Understand about security, SCM, CRM.

Credits: **4**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 3-1-0

Unit	Topic	No. of Lectures
I	Introduction to E-Commerce: Definition of E-Commerce, Scope of E-Commerce, driving forces for E-Commerce, Issues in implementing E-Commerce, E-Commerce Applications, Advantage- Business & Customer, Disadvantages, framework for understanding e-business, Classification of E-commerce, E-commerce Business Models.	15
II	Firewall: Types, Features, Characteristics, Electronic Data Interchange (EDI): Concept, Components, Differences between traditional EDI & Paper EDI, Advantages of EDI, Business Application of EDI, EDI Communication Process, EDI Security, Digital Signature.	15
III	Electronic Payment System: EPS Models, EPS Processing, Digital token based, debit card, smart card, Credit Card, risk in electronic payment system, E-auction: Introduction, Overview, Electronic trading, Online Banking: origin, advantages, disadvantages, Services.	15
IV	Web Security factors, E-Commerce security threats, security schemes, Protocols, Digital Certificates, Cyber law in India, Supply Chain Management (SCM): Components and issues, Customer Relationship Management (CRM): definition, Components, Benefits, ECRM: concept, impact, ECRM v/s CRM.	15

Suggested Readings:

- E-Commerce – Bharat Bhaskar
- E-Commerce -- CSV Murthy

Suggested online links:

- https://onlinecourses.swayam2.ac.in/cec22_mg05/preview

Programme/Class: **Degree**	Year: **Third**	Semester: **Sixth**
Subject: **Computer Science**		
Course Code: **CSCC363P**	Course Title: **Lab: Problem solving using Python**	

Course outcomes:

- To learn and understand Python programming basics.
- To learn and understand python looping, control statements and string manipulations.
- Students should be made familiar with the concepts of GUI controls and designing GUI applications.
- To learn and know the concepts of file handling, exception handling and database connectivity.

Credits**: 4**	**Core Compulsory**
Max. Marks: **30+70**	
Total No. of Lectures-Tutorials-Practical (in hours per week): 0-0-4	

List of Practicals:-

- Write a menu driven program to convert the given temperature from Fahrenheit to Celsius and vice versa depending upon user's choice.
- WAP to calculate total marks, percentage and grade of a student. Marks obtained in each of the three subjects are to be input by the user. Assign grades according to the following criteria:
- Grade A: Percentage >=80
- Grade B: Percentage>=70 and <80 Grade C: Percentage>=60 and <70 Grade D: Percentage>=40 and <60
- Grade E: Percentage<40
- Write a menu-driven program, using user-defined functions to find the area of rectangle, square, circle and triangle by accepting suitable input parameters from user.
- WAP to display the first n terms of Fibonacci series.
- WAP to find factorial of the given number.
- WAP to find sum of the following series for n terms: 1 – 2/2! + 3/3! ---------- n/n!
- WAP to calculate the sum and product of two compatible matrices.
- Programs of LIST
- Programs related to string manipulation
- Importing and executing built-in functions from the time, math and random modules
- Programs using break and continue statements.
- Programs related to dictionaries
- Programs using list comprehensions and anonymous functions
- Programs using the built-in methods of the string, list, tuple and dictionary
- Programs of Class and Object
- Programs of File Handling and Exception Handling

Suggested Readings:

- Allen B. Downey, “Think Python: How to Think Like a Computer Scientist”, 2nd edition, Updated for Python 3, Shroff/O’Reilly Publishers, 2016
- (http://greenteapress.com/wp/thinkpython/)
- Guido van Rossum and Fred L. Drake Jr, “An Introduction to Python – Revised and updated for Python 3.2, Network Theory Ltd., 2011.
- Charles Dierbach, “Introduction to Computer Science using Python: A Computational Problem-Solving Focus, Wiley India Edition, 2013.
- John V Guttag, “Introduction to Computation and Programming Using Python’’, Revised and expanded Edition, MIT Press , 2013
- Kenneth A. Lambert, “Fundamentals of Python: First Programs”, CENGAGE Learning, 2012.

Programme/Class: **Bachelor Research**	Year: **Fourth**	Semester: **Seven**
Subject: **Computer Science**		
Course Code: **CSCC471T**	Course Title: **Research Methodology**	

Course outcomes: After the completion of the course the students will be able to:

- Understand research methodology.
- Understand methods and techniques of data collection.
- Implement hypothesis.
- Prepare research report.

Credits: **4**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 3-1-0

Unit	Topic	No. of Lectures
I	Introduction to RM; Meaning, Objectives, Categories of research, Features of good research studies, types of research studies, Scientific and non-scientific methods, Research methods & research methodology, process of research. Importance of Research in Management Decisions, Defining Research Problems, Research design, Types of research design, research design in case of different research studies.	15
II	Methods and techniques of data collection; Types of data collection. Methods used for collection of different data types. Sampling and sampling distribution: importance of sampling, methods of sampling, sampling errors, Attitude measurement and Scales, Introduction to attitude- various methods to measure attitude	15
III	Data presentation, preparation and preliminary analysis, Concept, Sources and Types of Hypotheses; Formulation of Hypothesis; criteria of a good hypothesis, types of research hypothesis, advantages of Hypothesis, Hypothesis Testing Procedure, Type I and II Error, Chi-square test (Theoretical concepts & very simple numerical problems only)	15
IV	Report writing and presentation, Writing and Formatting of reports, Essentials of a Good Research Report, Presenting your research reports & tables, Graphical Presentation.	15

Suggested Readings:

- C.R Kothari : Research Methodology
- Dr. Kumar : Research Methodology

Suggested online links:

- https://onlinecourses.nptel.ac.in/noc22_ge08/preview

Programme/Class: **Bachelor Research**	Year: **Fourth**	Semester: **Seven**
Subject: **Computer Science**		
Course Code: **CSCC472T**	Course Title: **Cyber Security**	

Course outcomes: After the completion of the course the students will be able to:

- Understand types of information, cyber threats, and national/international cyber security standards.
- Do mathematical modelling and development of security techniques and information system.
- Develop understanding of legal issues related to cyber security.
- Apply ethical principles/responsibilities in cyber practices.

Credits: **4**	**Core Compulsory**

Max. Marks: 3**0+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 3-1-0

Unit	Topic	No. of Lectures
I	Introduction to Network Security, Aspects of Network Security, Need For Security, Goals Of Network Security, Features of a good Security policy, security attacks, security services and mechanisms, network security, Firewall: type, hardware and software, design principles.	15
II	Encryption/decryption, conventional encryption model, conventional encryption algorithms: character level encryption, bit-level encryption, private key encryption, Data encryption Standard algorithm, strength of DES.	15
III	Public key encryption, principles of public key cryptography systems, RSA algorithm, security of RSA, Fermat's and Euler's theorem, primality, The Chinese remainder theorem, Digital signature, application for public key cryptosystems.	15
IV	E-mail security, PGP, S\MIME security, S\MIME functionality, cryptographic algorithms, IP Security: application of IPsec, Benefits of IPsec, IPsec architecture, IPsec Services, Authentication header, Encapsulating Security payload. Web Security: threats on web, SSL and Transport layer Security. System security : virus and Intruders.	15

Suggested Readings:

- Michael E. Whitman and Herbert J. Mattord, "Principles of Information Security," Sixth Edition, Cengage Learning, 2017.
- Douglas J. Landoll, "Information Security Policies, Procedure, and Standards: A Practitioner's Reference," CRC Press, 2016.
- Harold F. Tipton, and Micki Krause, "Hand book of information security management," Sixth Edition, Archtech Publication, 2007.
- William Stallings, "Cryptography and Network Security: Principles and Practice," Sixth Edition, Pearson, 2014.

Suggested online links:

- https://onlinecourses.swayam2.ac.in/nou19_cs08/preview
- https://onlinecourses.swayam2.ac.in/cec22_mg05/preview

Programme/Class: **Bachelor Research**	Year: **Fourth**	Semester: **Seven**
Subject: **Computer Science**		
Course Code: **CSCC473T**	Course Title: **Artificial Intelligence**	

Course outcomes: After the completion of the course the students will be able to:

- Understand the basics of Artificial Intelligence and gain knowledge of the learning process and its models.
- Understand different types of search techniques.
- Understand different knowledge representation schemes.
- Understand the AI applications in the design of expert systems.

Credits: **4**	**Core Compulsory**

Max. Marks: 3**0+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 4-0-0

Unit	Topic	No. of Lectures
I	Introduction, The importance of AI , Applications of AI , general issue in AI problem solving , production system , knowledge: Definition & importance, knowledge based systems , components of knowledge based system , intelligent agents.	15
II	Search and Control Strategies, Examples of Search Problems(The eight puzzle, Travelling Salesman Problem), Uninformed Search (Breadth First Search , Depth First Search , Depth First Iterative Depending Search , Bidirectional Search) and informed search (Heuristic Information , Hill Climbing Methods ,Best First Search , Branch and Bound Search ,A* Search) , Searching AND-OR Graphs, The AO* Algorithm , Game Playing : mini max search approach.	15
III	Formalized Symbolic logics: Propositional logic , Syntax and Semantics for Propositional Logic ,Limitation of Propositional Logic , First order predicate Logic , Syntax and Semantics for FOPL ,Properties of Well Formed Formula , skolemisation, Conversion to Clausal Form , inference rules , unification, resolution principle.	15
IV	Semantic networks, frame system, value inheritance scripts, LISP and other AI programming languages, Introduction to Data Analytics: Sources and nature of data, classification of data (structured, semi-structured, unstructured), characteristics of data, introduction to Big Data platform, need of data analytics, evolution of analytic scalability, analytic process and tools, analysis vs reporting, modern data analytic tools, applications of data analytics.	15

Suggested Readings:

- Michael E. Whitman and Herbert J. Mattord, "Principles of Information Security," Sixth Edition, Cengage Learning, 2017.
- Douglas J. Landoll, "Information Security Policies, Procedure, and Standards: A Practitioner's Reference," CRC Press, 2016.
- Harold F. Tipton, and Micki Krause, "Hand book of information security management," Sixth Edition, Archtech Publication, 2007.
- William Stallings, "Cryptography and Network Security: Principles and Practice," Sixth Edition, Pearson, 2014.

Suggested online links:

- https://onlinecourses.swayam2.ac.in/nou19_cs08/previ ew

Programme/Class: **Bachelor Research**	Year: **Fourth**	Semester: **Seven**
Subject: **Computer Science**		
Course Code: **CSCC474T**	Course Title: **Web Designing**	

Course outcomes: After the completion of the course the students will be able to:

- Understand best technologies for solving web client/server problems.
- Analyze and design real time web applications.
- Use java script for dynamic effect and to validate from input entry.

Credits: **4**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 3-1-0

Unit	Topic	No. of Lectures
I	Internet evolution of internet, internet application, client and servers, hosts and nodes, internet services, different types of connections, dial up, leased, VSAT, internet service providers, choosing an ISP, DNS system, E-mail, basics e-mail functions, E-mail Protocols.	15
II	Introduction *to* HTML-- a short history of World Wide Web ,creating first HTML document, understanding basics of HTML tools, entering tags and attributes, applying structures tags, linking documents, URL, types of URL's, constructing link anchors, inserting e-mail, links, including images, developing images, adding images, using images as links, using background images.	15
III	Tables, creating basic tables, spanning rows and columns, adding captions, formatting tables, adding and formatting borders.HTML forms, developing forms, creating forms, Frames: creating frames, accommodating non-framed browser.	15
IV	Java Script: adding java script, what is java script, adding event handlers, Loops, control statements, functions, validations, Introduction to DHTML, Style Sheets.	15

Suggested Readings:

- Jeffrey C. Jackson, "Web Technologies: A Computer Science Perspective", Prentice Hall, 2007
- JavaScript: The Good Parts by Douglas Crockford
- HTML5 for Web Designers by Jeremy Keith
- The Art and Science of CSS: Create Inspirational, Standards-Based Web Designs by Cameron Adams
- Headfirst PHP & MySQL by Lynn Beighley & Michael Morrison

Suggested online links:

- https://onlinecourses.swayam2.ac.in/aic20_sp32/preview
- https://nptel.ac.in/courses/106/105/106105084/
- https://onlinecourses.swayam2.ac.in/aic20_sp11/preview

Programme /Class: **Bachelor Research**	Year: **Fourth**	Semester: **Seven**
Subject: Computer Science		
Course Code: **CSCC475P**	**Course Title: Lab: Web Designing**	

Course outcomes:

- Understand best technologies for solving web client/server problems.
- Analyze and design real time web applications.
- Use java script for dynamic effect and to validate from input entry.

Credits**: 4**	**Core Compulsory**
Max. Marks: **30+70**	
Total No. of Lectures-Tutorials-Practical (in hours per week): 0-0-4	

List of Practicals :-

1. Design a home page which displays information about your college department using headings, HTML entities and paragraphs.
2. Implement different type of list tags in the college department homepage.
3. Create a webpage for any clinic using marquee and HTML formatting tags.
4. Create 3 Hyperlinks in home page connecting it to 3 different pages.
5. Create 3 hyperlinks in a page, which jumps to 3 different headings on same page.
6. Insert image(s) and iframe in a webpage.
7. Design a page with image of block diagram of computer, mapping each component as area with specific co-ordinates which when clicked may give their detail.
8. Create a web page having two frames, Frame 1 containing links and another with contents of the link. When link is clicked appropriate contents should be displayed on Frame 2.
9. Design a timetable and display it in tabular format.
10. Demonstrate difference between "get" and "post" method of form tag in a form with name and password text fields.
11. Design an admission form for any course in your college with text, password fields, drop-down list, check-boxes, radio buttons, submit and reset button etc.
12. Create a website for online book store with Home, Login, Catalogue, Registration page with links to all these pages in a menu on top of every page. Embed heading, paragraph, images, video, iframe, form controls, table, list in this website.
13. Write a CSS style specification rule that would make all unordered lists (<ul> tags) have square bullets and a purple background.
14. Create a HTML form with the use of cascading style sheets.
15. Design a web page of your Home town with a attractive background color, text color, an image, font face by using Inline CSS formatting.
16. Create a catalog for an online shopping company that sells music records using style sheets.
17. Create a sample code to illustrate the External style sheet for your web page
18. Design a web page by using different CSS border styles.
19. Demonstrate the use of CSS Box Model.
20. Write a JavaScript program to display the current day and time .
21. Write a JavaScript program to get the current date.
22. Write a JavaScript program to find the area of a triangle.
23. Write a JavaScript program to determine whether a given year is a leap year.
24. Write a JavaScript program to calculate multiplication and division of two numbers.
25. Write a JavaScript program to convert temperatures to and from Celsius,
26. Write a JavaScript program to check whether a given positive number is a multiple of 3.

Suggested Readings:

- Jeffrey C. Jackson, “Web Technologies: A Computer Science Perspective”, Prentice Hall, 2007
- JavaScript: The Good Parts by Douglas Crockford
- HTML5 for Web Designers by Jeremy Keith
- The Art and Science of CSS: Create Inspirational, Standards-Based Web Designs by Cameron Adams.
- Headfirst PHP & MySQL by Lynn Beighley & Michael Morrison.

<table>
<tr><td>Programme /Class: Bachelor Research</td><td>Year: Fourth</td><td>Semester: Seven</td></tr>
<tr><td colspan="3">Subject: Computer Science</td></tr>
<tr><td>Course Code: CSCC476I</td><td colspan="2">Course Title: Internship</td></tr>
<tr><td colspan="3">Course outcomes:
• Understand best technologies for solving real time problems.
• Analyze and design real time web applications.</td></tr>
<tr><td>Credits: 4</td><td colspan="2">Core compulsory</td></tr>
<tr><td colspan="3">Max. Marks: 30+70</td></tr>
<tr><td colspan="3">Total No. of Lectures-Tutorials-Practical (in hours per week): 0-0-4</td></tr>
</table>

Programme/Class: **Bachelor Research**	Year: **Fourth**	Semester: **Eight**
Subject: **Computer Science**		
Course Code: **COM481T**	Course Title: **Data Analytics**	

Course outcomes: After the completion of the course the students will be able to:

- Describe the life cycle phases of Data Analytics through discovery, planning and building.
- Understand and apply Data Analysis Techniques.
- Implement various Data streams.
- Understand item sets, Clustering, frame works & Visualizations.

Credits: **4**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 3-1-0

Unit	Topic	No. of Lectures
I	Introduction to Data Analytics: Sources and nature of data, classification of data (structured, semi-structured, unstructured), characteristics of data, introduction to Big Data platform, need of data analytics, evolution of analytic scalability, analytic process and tools, analysis vs reporting, modern data analytic tools, applications of data analytics. Data Analytics Lifecycle: Need, key roles for successful analytic projects, various phases of data analytics lifecycle – discovery, data preparation, model planning, model building, communicating results, operationalization.	15
II	Data Analysis: Regression modelling, multivariate analysis, Bayesian modelling, inference and Bayesian networks, support vector and kernel methods, analysis of time series: linear systems analysis & nonlinear dynamics, rule induction, neural networks: learning and generalisation, competitive learning, principal component analysis and neural networks, fuzzy logic: extracting fuzzy models from data, fuzzy decision trees, stochastic search methods	15
III	Mining Data Streams: Introduction to streams concepts, stream data model and architecture, stream computing, sampling data in a stream, filtering streams, counting distinct elements in a stream, estimating moments, counting oneness in a window, decaying window, Real-time Analytics Platform (RTAP) applications.	15
IV	Frequent Item sets and Clustering: Mining frequent item sets, market based modelling, Apriority algorithm, handling large data sets in main memory, limited pass algorithm, counting frequent item sets in a stream, clustering techniques: hierarchical, K-means, clustering high dimensional data, CLIQUE	15

	and Proclus, frequent pattern based clustering methods, clustering in non-Euclidean space, clustering for streams and parallelism.	

Suggested Readings:

- Michael Berthold, David J. Hand, Intelligent Data Analysis, Springer
- Anand Rajaraman and Jeffrey David Ullman, Mining of Massive Datasets, Cambridge University Press.
- Bill Franks, Taming the Big Data Tidal wave: Finding Opportunities in Huge Data
- Streams with Advanced Analytics, John Wiley & Sons.

Suggested online links:

- https://epgp.inflibnet.ac.in/Home/ViewSubject?catid=fBYc kQ KJvP3a/8Vd3L08tQ==

Programme/Class: **Bachelor Research**	Year: **Fourth**	Semester: **Eight**
Subject: **Computer Science**		
Course Code: **COM482T**	Course Title: **Software Testing**	

Course outcomes: After the completion of the course the students will be able to:

- Understand about Software Testing.
- Understand about different testing Techniques.
- Implement Test Cases.
- Understand about the parameters and metrics of software testing.

Credits: **4**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 4-0-0

Unit	Topic	No. of Lectures
I	Testing Software:Software Testing and its importance, Module Testing, Integration Testing, Top-down versus bottom up testing, Mixed testing. Software Faults and Failures: Types of faults, Origins of Software Defects, The Cost of Repairing Defects, Verification and Validation :Verification Testing, Coding standards, Walk-Through, Formal Inspection, Verifying . Requirements, Verifying Functional Design, Validation Test Criteria, Design metrics.	15
II	Testing Techniques and Strategies: White-Box Testing: Flow graph notation, Cyclomatic Complexity, Control Structure and Loop Testing, Dataflow Testing.Black-Box Testing: Graph-based testing methods, Equivalence partitioning, Boundary Value Analysis ,UNIT Testing, Integration Testing, System Testing, Exhaustive Testing.	15
III	Building Tests and Test Plans: Designing and Creating Tests, Maintaining Checklists, White-box Test Cases and Test Procedures, Test Data Selection and Outputs, Black-box test cases and test procedures, Planning and Creating Test Plans. Testing Specialized Systems and Applications: Graphical User Interface (GUI) Testing, Usability Testing, Client/Server Architectures and Web Testing, Testing OO Systems, Volume and Stress Testing Testing Measurements: Software Size and Complexity, Function Point Analysis.	15
IV	Quality Assurance and Standards: Quality and Quality Assurance (QA), Techniques of quality Assurance, Software Testing and QA,Software Development Models, Quality metrics, Configuration	15

	Management, Quality management models *(ISO,* SPICE, IEEE, and CMM).	

Suggested Readings:

- W.M. Perry, “Effective Methods for Software Testing", 2002, Wiley.
- Nina Godbole, “Software Testing and Quality Management”
- A. Behforooz and F.Hudson, “Software Engineering Fundamentals", 1996, OUP.
- James Peters & W. Pedrycz, "Software Engineering", 2000, Wiley.
- S. Pfleeger, "Software Engineering", 2001 , Peterson Education.

Suggested online links:

https://onlinecourses.nptel.ac.in/noc22_cs61/preview

Programme/Class: **Bachelor Research**	Year: **Fourth**	Semester: **Eight**
Subject: **Computer Science**		
Course Code: **COM483T**	Course Title: **Web Development**	

Course outcomes: After the completion of the course the students will be able to:

- Understand best technologies for solving web client/server problems.
- Analyze and design real time web applications.
- Use Java script for dynamic effect and to validate from input entry.

Credits: **4**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 3-1-0

Unit	Topic	No. of Lectures
I	Web Programming Introduction ,Understanding role of Web Server and Web Browser. Brief about HTTP Protocol. HTTP Request structure. Form Tag and comparison between Get and Post methods. HTTP Response Structure HTML and JavaScript Understanding HTML Form Tag and elements within it. Javascript using Sample Programs. Traditional ASP Basics Introduction to ASP. How ASP Works. ASP Objects. Installing IIs on Windows 7 & Windows 8. Sample Programs. Importance's of Form tag and how it works.	15
II	ASP.NET Introduction & Controls ASP.NET Introduction First ASP.NET Application. Auto Postback Property. Event Handler Parameters. Dynamically intializing Controls IsPostBack property of Page class ListControls Comparison between HtmlControls and WebControls Control Properties and Methods FileUpload Control ASP.NET Architecture What is AppDomain? Life Cycle of ASP.NET Page How Control Manages its State What is EnableViewState How Control raises events Event handling in Web Forms Writing Custom Classes in WebApplication CSS & Themes Work with CSS Use Themes to Customize a Site Name Skins within a Theme Do server-side Styles using Themes Add contents of a Theme and Skin Apply themes and Profiles	15
III	Redirecting User to Another Page Redirecting Options Response.Redirect Server.Transfer Cross Page Postback Determine how the pages was invoked Passing values between pages Master Pages Introduction to MasterPage ContentPlaceHolder and Content tags Accessing controls of MasterPage in ContentPage URL's in MasterPages UniqueID and ClientID Working with User Control Overview of User Controls Creating a User Control Adding Properties to User Control Adding Events to User Control Using User Control in Web Form Rendering Clients Scripts Using Page.ClientScript methods Validation Controls Base Validator Required Field Validator Compare Validator	15

	Range Validator Regular Expression Validator Custom Validator Causes Validation Property	
VI	Grouping - Validation Group Property Page.Validators and Page.IsValid ASP.NET State Management Static Members View State Hidden Field in Form Query String HttpContext Cookies-HttpCookie Sessions-HttpSessionState Application-HttpApplicationState Summary of All Features SqlDataSource GridView DetailsView FormView DataList Repeater ListView AJAX.NET .	15

Suggested Readings:

- Beginning ASP.NET 2.0 with C# (Wrox Beginning Guides)

Suggested online links:

- https://www.tutorialspoint.com
- https://www.guru99.com
- https://asp.net-tutorials.com

Programme/Class: **Bachelor Research**	Year: **Fourth**	Semester: **Eight**
Subject: **Computer Science**		
Course Code: **COM484P**	Course Title: **Lab: Web Development**	

Course outcomes: After the completion of the course the students will be able to:

- Understand best technologies for solving web client/server problems.
- Analyze and design real time web applications.
- Use Java script for dynamic effect and to validate from input entry.

Credits: **4**	**Core Compulsory**

Max. Marks: **30+70**

Total No. of Lectures-Tutorials-Practical (in hours per week): 0-0-4

List of Programs:

1. Implementation of Mandatory vs. optional fields in forms.
2. Create mailing list forms in asp.net
3. Implementation of default values in web forms.
4. How to apply input masking in asp.net web forms.
5. How to make form keyboard-friendly.
6. How to set autofocus for input field.
7. How make forms with auto completion in asp.net.
8. How to implement sentence case and title case and avoid all caps.
9. How to upload documents in asp.net web forms.
10. How to implement validation in web forms.

Programme /Class: **Bachelor Research**	Year: **Fourth**	Semester: **Eight**
Subject: **Computer Science**		
Course Code: **CSCC485R**	Course Title: **Research Project**	
Course outcomes: • Understand best technologies for solving real time problems. • Analyze and design real time web applications.		
Credits**: 12**	**Core Compulsory**	
Max. Marks: **300**		
Total No. of Lectures-Tutorials-Practical (in hours per week): 0-0-12		

Syllabus of Minor Subjects

Programme/Class: **Certificate**	Year: **First**	Semester: **First**
Subject: **Computer Science**		
Course Code: **CSMC11T**	Course Title: **: Information Technology**	

Course outcomes:

- To understand Functional Components of Computers.
- Effectively explains the basic concepts of Hardware and software
- Able to understand Operating system, Functions of Operating System.
- Able to get overview of Data Communication and Internet

Credits: 4	**Minor Elective**

Unit	**Topic**	**No. of Lectures**
I	Introduction, Characteristics of Computers, Block diagram of computer, Types of computers and features: Mini Computers, Micro Computers, Mainframe Computers, Super Computers. Types of Programming Languages: Machine Languages, Assembly Languages, High Level Languages, Translators:- Assembler ,Compiler, Interpreter and Linkers, Operating system concepts, Types of OS, Functions of OS	15
II	I/O Devices:- Keyboard, Mouse, Scanner, Light pen, Trackball, Joystick, Bar NUMBER reader, OCR, OMR, MICR, Digitizer; Monitor, Printer, Plotter etc., Memory concepts, Types of Memory (Primary and Secondary):-RAM,ROM and its types, Secondary Storage Devices (Magnetic tape, Magnetic Disk(FD, HD),Optical Disk(CD,DVD), Pen drive.Data Organization: Drives, Files, Directories	15
III	Basic elements of a communication system, Data transmission modes, Data Transmission speed, Data transmission media ,Digital and Analog Transmission, Network topologies, Network Types (LAN, WAN and MAN), Client and Servers , Intranet, Extranet.**Internet:** Introduction to Internet, Terminologies related to Internet: Protocol, Domain name, IP address, URL, World Wide Web, Connecting to the Internet	15
IV	Introduction to Big Data: Types of digital data, Big Data architecture and characteristics, Big Data technology components, Big Data importance and applications, Introduction to Block Chain: Structure, Operational aspects of Bitcoin Block, Compare different types of Block chains, Protocols, Payment mode code execution.	15

Suggested Readings:

- P. K. Sinha & Priti Sinha, "Computer Fundamentals", BPB Publications, 2007.
- Dr. Anita Goel, Computer Fundamentals, Pearson Education, 2010.
- Fundamental of Computers – By V. Rajaraman B.P.B. Publications

Suggestive online links:

https://www.pearsoned.co.in/prc/book/anita-goel-computer-fundamentals-1e1/9788131733097

Programme /Class: **Diploma**	Year: **Second**	Semester: **Third**
Subject: **Computer Science**		
Course Code: **CSMC31T**	Course Title: **Principles of Organizational Theories**	

Course outcomes: After the completion of the course the students will be able to:

1. Understand the fundamentals of Organizational Behaviour (OB) - such as working with people, nature of organizations, leadership and motivation of people.
2. To help students develop a conceptual understanding of Organizational Behaviour theories.
3. To enable the students to put the ideas and skills of Organizational Behaviour into practice.

Credits: **4**	**Minor Elective**

Unit	Topic	No. of Lectures
I	Introduction to organizational behaviour: nature, scope, challenges of organizational behaviour [5], managing people and organization [5], organizational behaviour in global context organizational behaviour model [5].	15
II	Work teams - concept and types. Team effectiveness. Team creation. Power-bases of power, contingency approach to power [5]. Politics -concept, causes and management, leadership concept, styles, [5] theories- Trait theory, charismatic leadership theory, behavioural theory [5].	15
III	Organization theory - classical, neoclassical and modern [5]. Organizational culture concept impact, creating organization culture [5]. Organizational design-importance types of OD, consequences of poor OD [5].	15
IV	Organizational change-factors, planned change, change resistance to change and overcoming resistance [5], Organization development- definition, process of organizational development. [5] Action research model, Organizational development interventions-sensitivity training and organizational development [5].	15

Suggested Readings:

- K. Aswathappa, “Organizational behaviour”, Himalaya Publishing House.
- Kavita Singh: Organisational Behaviour 3rd Ed. Vikas Publication.
- Robbins Stephen P and Judge T.A., Vohra, Organisational Behaviour, 16th Ed. Pearson. [Chapter 9,12,13,14,17]
- Stephen P. Robbins, Jennifer George and Gareth Jones, “Management and Organizational Behaviour”, Pearson Education Inc.

Syllabus of Vocational Subjects

Programme /Class: **Certificate**	Year: **First**	Semester: **First**
Subject: **Computer Science**		
Course Code: CSVC11	Course Title: **Office Automation I (MS Word and MS Powerpoint)**	

Course outcomes:

- Ms. Office course trains students how to use MS Office applications use in office work such as creating professional-quality documents;
- Store, organize and analyze information;
- Arithmetic operations and functions; and
- Create dynamic slide presentations with animation, narration, images, and much more, digitally and effectively

Credits: **3**	**Vocational**

Unit	Topic	No. of Lectures
I	Microsoft Word: Creating Word Documents with Headers and Footers, Formatting Documents with Themes and Styles , Editing and Correcting Documents , Adjusting Alignment and Spacing, Formatting Text, Improving a Document with Find and Replace and AutoCorrect, Checking Spelling and Grammar	12
II	Working with Table, Inserting Pictures, Text Boxes, and Shapes, Formatting Graphics Objects , Working with SmartArt Graphics, Text Effects, and Page Borders, Managing customer lists and creating mailings.	12
III	Microsoft PowerPoint: Working with Slides, Managing content on slides, Working with Headers, Footers, and Notes, Inserting and Formatting Pictures, Formatting Text, Aligning Text, Displaying the Presentation Outline.	11
IV	Arranging Slides, Adding Slide Transitions, Adding slide animations, organizing and printing presentations, Configuring slideshows, Managing slide masters and presentation settings.	10

Suggested Readings:

- Microsoft Office 2007 Bible - John Walkenbach,Herb Tyson,Faithe Wempen,cary N.Prague,Michael R.groh,Peter G.Aitken, and Lisa a.Bucki -Wiley India pvt.ltd.
- A Conceptual Guide to OpenOffice.org 3 - R. Gabriel Gurley- CreateSpace Independent Publishing Platform, 2008

Suggested online links:

- https://wiki.openoffice.org/wiki/Documentation

Programme /Class: **Certificate**	Year: **First**	Semester: **Second**
Subject: **Computer Science**		
Course Code: CSVC12	Course Title: **Office Automation II (MS Excel and MS Access)**	

Course outcomes: After the completion of the course the students will be able to:

- How to use MS Office applications use in office work such as creating professional-quality documents.
- Store, organize and analyze information in spreadsheet.
- Arithmetic operations and functions.
- Create database and perform queries.

Credits: **3**	**Vocational**

Unit	Topic	No. of Lectures
I	Create your first Excel file, enter data and create a table, save and open Workbooks, Working with multiple Worksheets, Moving around a spreadsheet, Copying, pasting, inserting and deleting cells, ranges of cells, rows, columns and Worksheets. Grouping rows and columns; changing cells size and shape, Format cells by entering specific data, selecting fonts and color, Merging of cells, Cell Naming.	12
II	Data Validation, Cell Referencing: Relative, Absolute and Mixed, Understanding Excel Formula, copying formulas, Popularly used Excel functions, Formula Auditing, Sorting, Filtering: Auto Filter and Advanced Filter.	12
III	**MS Access:** Introduction, RDBMS, Data Types, Create Database, Planning a Database, Starting Access, Creating a New Database, Creating Tables, Adding Primary Key and Constraint.	11
IV	Query Data, Query Criteria, Action Queries, Create Queries with Query Wizard, Create Relationships, One-To-One Relationship, One-To-Many Relationship, Many-To-Many Relationship, Work with Forms, Sort, Retrieve, Analyze Data	10

Suggested Readings:

- Microsoft Office 2007 Bible - John Walkenbach,Herb Tyson,Faithe Wempen,cary N.Prague,Michael R.groh,Peter G.Aitken, and Lisa a.Bucki -Wiley India pvt.ltd.
- A Conceptual Guide to OpenOffice.org 3 - R. Gabriel Gurley- CreateSpace Independent Publishing Platform, 2008

Suggested online links:

https://wiki.openoffice.org/wiki/Documentation

Programme /Class: **Diploma**	Year: **Second**	Semester: **Third**
Subject: **Computer Science**		
Course Code: CSVC21	Course Title: **ICT Tools and its Applications[Vocational Subject]**	

Course outcomes: After the completion of the course the students will be able to:

- Students will be able to know about the Basics of Gmail account creation
- Know about the Basics of Search Engine, video uploading and downloading
- Know Different Social Media strategies and basics of Blogging.

Credits: **3**	**Vocational**

Unit	Topic	No. of Lectures
I	How to create Gmail account, Strategy to manage google drive: Creating folders, organizing folders, uploading files and folders on google drive, how to create Google Form and Quiz, Manage Responses in Google sheet, Google Classroom: creating and join class, sharing files and classwork in classroom.	12
II	What is search Engine, Search Engine Basics, Major Search Engines, searching using keywords, Advanced searching. YouTube Profile, how to Create YouTube Channel, Uploading Video and optimization, Increase Video Views, Increase subscriber, Social Sharing, Comments.	12
III	Introduction of Facebook, Social Media Strategy, setting up Facebook and Privacy, Facebook Profile, what can you do with Facebook, Facebook pages, events and groups. Introduction to Twitter, Twitters profile, Followers, Following, Tweeting, Twitter settings, how to increase followers and using # tag.	11
IV	What is blogger. What is blogging, how to create blog on BlogSpot, how to create blog on blog website, how to create a new post, how to create page, how to earn money from blogging, how to customize and manage setting, how to promote blog website.	10

Suggested online links:

- https://www.info.com/serp?q=create+gmail+account&segment=info.rc001&s1recircid=dl2LF2vOuVXArJJK19Zi&gclid=CjwKCAjw9J2iBhBPEiwAErwpeTEsDVT1Y6bqOFbt6-UlfB3AU6g0CuNhQUezj8KnLlKzR8nkVzmSFhoC4GcQAvD_BwE&utm_source=adwords&s1sid=0fobmd9cwtqbb049ehpcomj1
- https://www.techtarget.com/whatis/definition/search-engine
- https://help.twitter.com/en/managing-your-account/accessing-your-twitter-data
- https://www.wix.com/blog/2020/02/what-is-a-blog/?utm_source=google&utm_medium=cpc&utm_campaign=16242175905^134377093918&experiment_id=^^582523585386^^_DSA&gclid=CjwKCAjw9J2iBhBPEiwAErwpeSGn4MNc7DfbF28sMGAL0crtFrwtTqdz6Whi3LUDsS3lLzxrPdnXsRoCSdUQAvD_BwE

Programme /Class: **Diploma**	Year: **Second**	Semester: **Fourth**
Subject: **Computer Science**		
Course Code: CSVC22	Course Title: **Introduction to Web Designing**	

Course outcomes: After the completion of the course the students will be able to:

- Understand best technologies for solving web client/server problems.
- Analyze and design real time web applications.

Credits: **3**	**Vocational**

Unit	Topic	No. of Lectures
I	Internet evolution of internet, internet application, client and servers, hosts and nodes, internet services, different types of connections, internet service providers, choosing an ISP, DNS system, E-mail, basics e-mail functions, E-mail Protocols.	15
II	Introduction *to* HTML-- a short history of World Wide Web ,creating first HTML document, understanding basics of HTML tools, entering tags and attributes, applying structures tags, linking documents, URL, types of URL's, constructing link anchors, inserting e-mail, links, including images, developing images, adding images, using images as links, using background images.	15
III	Tables, creating basic tables, spanning rows and columns, adding captions, formatting tables, adding and formatting borders.HTML forms, developing forms, creating forms, Frames: creating frames, accommodating non-framed browser.	15
IV	Introduction to DHTML, Style Sheets., Cascading Style sheets,Various ways of adding style sheets, CSS properties: background, font, text, link, list, margin, border, padding,float, z-index	15

Suggested Readings:

- HTML5 for Web Designers by Jeremy Keith
- The Art and Science of CSS: Create Inspirational, Standards-Based Web Designs by Cameron Adams

Suggested online links:

- https://onlinecourses.swayam2.ac.in/aic20_sp32/preview
- https://nptel.ac.in/courses/106/105/106105084/
- https://onlinecourses.swayam2.ac.in/aic20_sp11/preview